Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 3rd Year 1st Term Examination, 2021

URP 3121

(Rural Development Planning-I)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Draw necessary diagrams if required.

Section – A

1. a) Distinguish between absolute and relative measures of poverty with necessary examples. (08)
b) Discuss various absolute measures of poverty with necessary example. (15)
c) Determine the followings for the given data (12)
i) Headcount Index ii) Poverty Gap Index iii) Square Poverty Gap Index

Expenditure for each individual in regions ($)				
Region-A	90	115	100	150
Region-B	105	120	95	150
Region-C	110	100	120	150

Also please explain the results assuming Poverty Line: 110$

2. a) What is meant by the basic need approach of poverty measurement? (08)
b) Discuss various non-monetary measures of rural development having implication on property. (15)
c) What is the Gini-coefficient? From the following data compare the income disparity level of Region A and Region B. (12)

Income (BDT)	Region A	Region B
	No. of Person	No. of Person
1000	600	1500
1200	800	1000
1400	1200	900
1600	900	1100
2000	1000	300
2300	500	200

3. a) Draw your concept on endogenous approach and participatory approach of rural development. (08)
b) Provide a brief account of the development programmes initiated in Bangladesh. (15)
c) What is meant by the Sound Safety Net program? Discuss the role of rural safety nets in rural poverty reduction in Bangladesh. (12)

4. a) Discuss important conditional and unconditional safety net programs of Bangladesh. (15)
b) Describe in brief the Ashrayan program of rural development in Bangladesh (10)
c) Discuss the "My Village My Town" initiatives of the government in light of its potential role in transforming the rural life and livelihood in Bangladesh. (10)

Section – B

5. a) Briefly explain the influencing factors of rural settlement pattern. (10)
   b) Discuss different settlement patterns with necessary sketches. (15)
   c) Discuss in brief the procedure of a hierarchical rural centre plan. (10)

6. a) Critically evaluate the main components of the Comilla Model in development of Bangladesh. (15)
   b) Briefly narrate the approach and ideas of rural development during the period 1950 to 2000 (10)
   c) Give a short description on the success of V-AID programme of rural development (10)

7. a) Define rural industries. (05)
   b) Make an overview about different types of rural Industries with appropriate examples. (15)
   c) Briefly discuss with example the use of industrial development screening matrix for selecting a suitable location of a rural industry. (15)

8. Write short notes on the following (any five): (5*7)=35
   a) Lorenz Curve
   b) Todaro's Migration Model
   c) RWP
   d) ~~FFWR~~ FFWP
   e) NGOs in rural development
   f) Rural Housing

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 3rd Year 1st term Regular Examination, 2021

CE 3151

(Fundamentals of Transportation Engineering)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume suitable data if necessary.

Section – A

1 (a) What are traffic zones? Write down the elements of transportation system and describe them in details. (12)

(b) Define transportation planning process. What are the objectives of transportation planning? (08)

(c) Explain the inter-relation of land use and transportation planning. Why are land use planning and control important in Transportation Engineering? (15)

2 (a) What is the prerequisite condition for transportation demand analysis? Enlist the differences between restrictive and strategic measures of transportation demand. (10)

(b) Define: i. ADT, ii. AADT, iii. AWT, iv. Monthly variation factor (08)

(c) Write down different methods for traffic volume survey. Suppose in a city, in 2012, total 15,800 vehicles passed a point of a road at the month February. If the total vehicles that passed that point of the road in that year 2012, were 0.3 million, find out the monthly variation factor of February in that year. (12)

(d) Define O-D survey. Write down different modes of transport. (05)

3 (a) How does a planner collect the O-D data from field? Write down the objectives of O-D survey. (10)

(b) What is parking survey? Write down the differences between on-street and off-street parking facilities. (10)

(c) Define parking index. What are the field-based survey of public transport survey? (05)

(d) What are the things that should be considered by a planner during forecasting the urban goods movement? Enlist the importance of infrastructure for economic growth. (10)

4 (a) What is pipeline transport? Write down the advantages and disadvantages of water transport. (10)

(b) What is highway capacity? Describe the types of roadway capacity. (12)

(c) What is parking study? Write down the different types of parking facilities. (13)

Section – B

5 (a) What is meant by "Highway Geometrics"? What are the elements of Highway Geometry? (05)

(b) What is kerbs? Classify kerbs based on functions and describe briefly. What are the uses of road kerbs? (15)

(c) Why is extra width of road provided on horizontal curve? (07)

(d) Draw neat sketch of a typical urban and rural road section with its major elements. (08)

6 (a) Define following terms: (08)

(i) Shoulder
(ii) Camber
(iii) Carriageway
(iv) Right of way

(b) What are the provisions for elderly and handicapped pedestrians? Why extra-widening is necessary on horizontal curve? (10)

(c) A horizontal curve is to be designed for a section of a highway having a design speed of 70 mi/h. (10)
Assume, fy = 0.15

(i) If the physical conditions restrict the radius of curve to 650 ft, what value is required for the super elevation at this curve?
(ii) Is it good to design?

(d) Why is camber provided in road surface? What are the factors which affect amount of camber? (07)

7 (a) What is meant by "Super-elevation"? Derive an expression for super-elevation at level road. (10)

(b) The design speed of a highway is 85 km/h. There is a horizontal curve of radius 145 m. Calculate the super-elevation needed to maintain this speed. If maximum super-elevation of 1 in 15 is not exceeded, calculate to the maximum allowable speed on this horizontal curve. (10)

(c) What is the form of vertical curves in road? Discuss briefly. (10)

(d) Why is extra widening provided on the horizontal curve? (05)

8 (a) What is sight distance? Discuss the different kinds of sight distance. (07)

(b) Derive an expression for finding overtaking sight distance at road. (10)

(c) Calculate the safe stopping sight distance for design speed of 80 km/h, for (10)

(i) Two-way traffic on a two lane road.
(ii) Two-way traffic on a single lane road.

Assume, $\mu = 0.38$ and reaction time = 2.5 sec.

(d) Discuss about the sight distance at intersection. (08)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 3rd Year 1st Term Examination, 2021

Hum 3117

(Accounting and Auditing for Planners)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. a) Describe the role of accounting in financial decision making and investment. (10)
b) Describe the principle of accounting with example. (15)
c) Accounting is the ultimate tool for business decisions- Argue and state your position. (10)

2. a) Define journal. Why is it called the book of original entry? (10)
b) Journalize the following transaction in the book of Mr. Peter: (25)

June 1: Mr. Peter invested cash 1,00,000 and a building of taka 5,00,000 to start the business.
June 5: Purchased and equipment of taka 50,000 on account making half of the amount down payment.
June 10: Appointed and assistant whose salary will be 20,000 taka per month
June 15: Made a pertial payment of taka 5,000 on the balance vote for equipment.
June 16: Collected taka 30,000 from the clients by providing services.
June 20: Billed clients with taka 20,000 for services provided.
June 25: Paid utilities bill taka 10,000.
June 30: Paid salary to the assistant for the month of June, taka 20,000.

3 a) Describe the limitations of trial balance. How these limitations can be addressed? (10)
b) What information can be extracted from cash flow statement about an organization? What are the uses of the information? (10)
c) Prepare necessary ledger accounts for the following transactions: (15)

June 01, 2022: Purchase goods in cash taka 25,000.
June 03: Paid for 12-month insurance premium taka 6,000.
June 15: Received from customers in advance taka 20,000.
June 21: Distributed dividend taka 15,000.
June 25: Receive interest in cash taka 5,000.

4 a) The following ledger balances are obtained from the books of account of Pioneer Consultant (Pvt) Ltd. for the year ended on June 30, 2020. (35)

Pioneer Consultant (Pvt.) Ltd.			
Trial balance as on June 30, 2020			
Accounts title	Debit (Taka)	Accounts title	Credit (Taka)
Cash	18,500	Loan payable	20,000
Bank	15,300	Accounts payable	13,500
Account receivable	17,000	Capital	42,000
Prepaid insurance	9,000	Service revenue	86,900
Supplies	12,500	Salaries payable	7,400
Building	71,000		
Drawings	4,000		
Utilities expenses	8,500		
Wages	5,300		
Interest	1,500		
Advertising expenses	7,200		
	1,69,800		1,69,800

Other data:

i. Depreciation charged on buildings @10% per year.
ii. Service provided but unbilled taka 5,200.
iii. Interest accrued on loan payable @9%.
iv. Insurance expired @500 per month.
v. Supplies expense of the year 8,500 taka.
vi. Bad debt calculated taka 3,500.

Required:

1) A statement of comprehensive income.
2) A statement of owner's equity.
3) A statement of financial positions.

Section – B

5. a) Define financial audit. Briefly describe the elements of a formal financial audit. (15)
   b) What is fundament financial reporting? Write down the cause that might instigate committing fraud. (15)
   c) Define value-for-money audit. (05)

6 a) What is an audit opinion? Discuss the kinds of audit opinions in short. (20)
  b) When does an auditor provide disclaimer of opinion and adverse opinion? (15)

7 a) What are the objectives of internal control system of an organization? (10)
  b) Describe the components of internal control. (20)
  c) What is the custody procedure of an internal control system (05)

8 a) What is 'Tax Holiday'? How do tax holiday contributes to government revenue at the time of needs? (10)
  b) Briefly describe the text practices of corporates and individuals in Bangladesh. What are the current tax brackets of individuals and corporates? (15)
  c) Explain the needs of taxation in public finance. (10)

KhulnaUniversity of Engineering & Technology

Department of Urban & Regional Planning

BURP 3rd Year 1st Term Examination, 2021

URP 3161

(Housing and Real Estate Development)

Full Marks: 210 | Time: 3.00 hrs

N.B. i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define the following terms: "House", "Housing", and "Adequate Housing". (12)
   (b) Explain "Maslow's hierarchy of needs". How do you put housing need in that hierarchy? Explain with examples. (15)
   (c) "Housing is a need, which can also become a demand". Explain this statement with example. (08)

2. (a) Classify housing based on formality, provision, distribution, typology, ownership and location. (20)
   (b) Write short notes on: (15)
   (i) Social housing
   (ii) Land banking
   (iii) Site and services

3. (a) What is perfect market competition? What are the common non-market factor that influence land price in Bangladesh? (13)
   (b) Distance from city center has inverse relationship with land price. Explain high land price around KUET campus even through high distance from Khulna City center. (12)
   (c) "Decadal change in the international housing policies followed a closed-loop"-explain. (10)

4. (a) What are the major sources of sectoral housing in Bangladesh? Which sectoral housing is the most important according to the National Housing Policy (present) of Bangladesh and why? (10)
   (b) What was the World Bank condition for site-and-services? How and why government site-and-services failed and which component was most significant for the failure? (10)
   (c) Singapore is a success story for preferential provision of public housing. What are the major characteristics of Singapore's housing approach? (15)

Section – B

5. (a) Explain the four combinations of real estate investment risk with examples. (20)
   (b) When will the downturn actually occur in a real estate market? Explain with supply and demand curve. How a developer will survive during a market downturn? (15)

6. (a) Market research approach focuses from broad to specific. Explain with example. (15)

   (b) Write short note on the following: (20)
   (i) Economic Base Multiplier
   (ii) Bid-Rent Model
   (iii) Circle Model
   (iv) Sector Model

7. (a) Discuss the primary principles of real estate valuation. (20)

   (b) What are the adjustments need to be considered for valuation of property using the traditional sales comparison approach? (15)

8. (a) "Real estate includes not only the land surface of the earth but also the permanents improvements above and below the surface."-Why? Explain with the example. (15)

   (b) True real estate decisions are about acquiring, financing, using, improving and disposing of actual real estate assets. Explain the statement. (20)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 3rd Year 1st Term Regular Examination, 2021

CE 3107

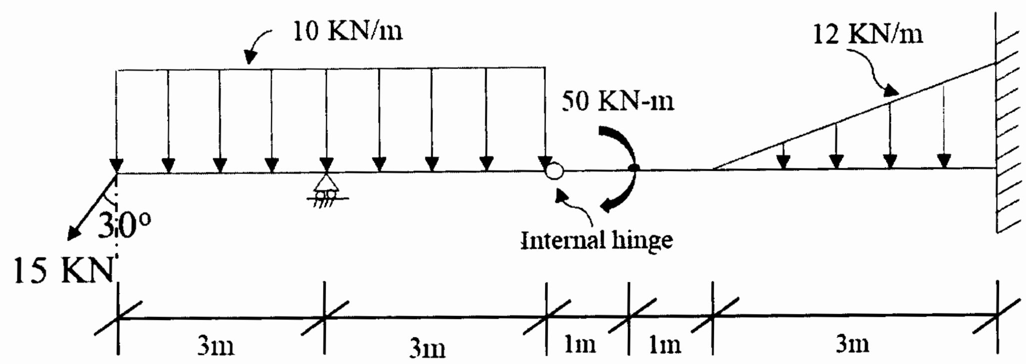
(Elements of Solid Mechanics)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume suitable value/data if necessary.

Section – A

1 (a) Draw the axial force, shear force and bending moment diagram for the following loaded beam. Also, write down the axial force, shear force and bending moment equations. (35)

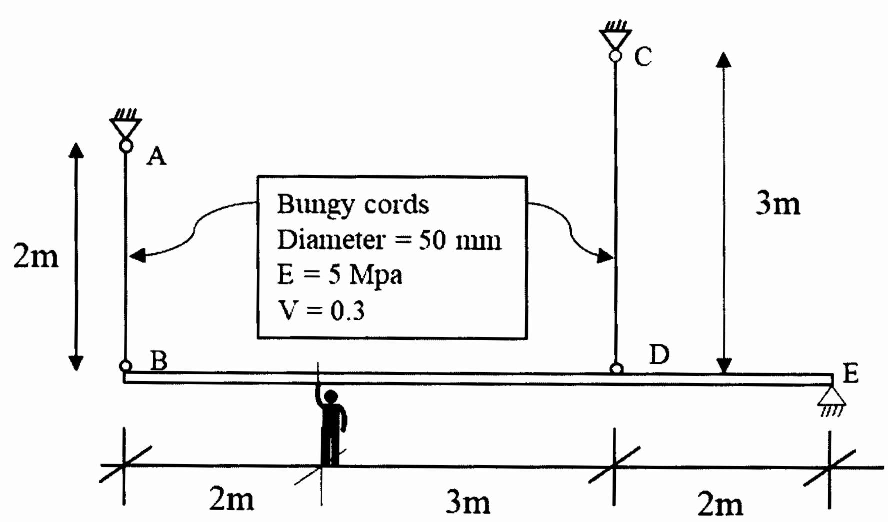


2 (a) Draw a typical stress-strain diagram of structural steel. How to determine the yield strength by offset method – Explain. (10)

(b) A horizontal bar of negligible mas is hinged at point E and suspended at B and D from bungy cords AB and CD. The cords AB and CD are connected to pin supports at A and C respectively. If you are hanging at point G with a mass of 60 kg (25)

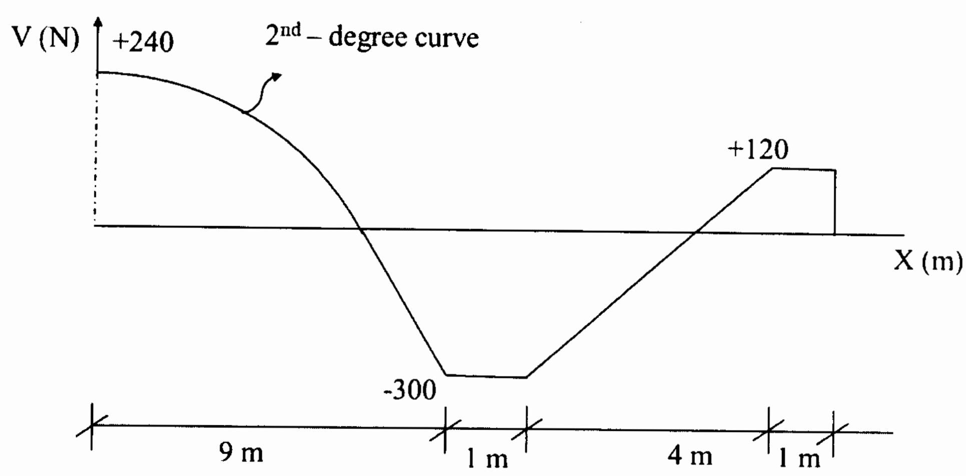
(i) What will be the deflection at point D?
(ii) Determine the support reaction at E
(iii) For bungy cord AB, determine the force, normal stress, normal strain and change in diameter.

Assume the horizontal bar is rigid and $g = 9.81\ ms^{-2}$.

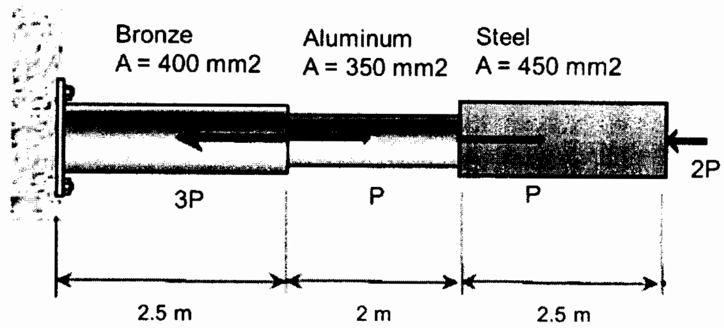


3 (a) Derive the relations among the loads, shears and bending moments in a beam. Write down the sign convention of shear force and bending moment. (15)

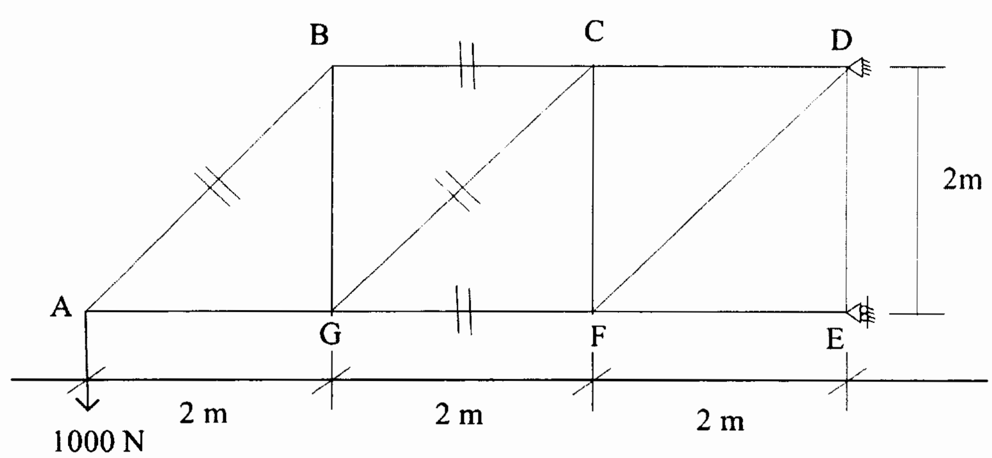
(b) Draw moment and load diagrams corresponding to the following shear diagram. (20)



4 (a) An aluminum road is rigidity attached between a steel rod and a bronze road as shown in figure below. Axial loads are applied at the position indicated. Find the maximum value of P that will not exceed a stress in steel of 150MPa, in aluminum of 90 MPa, or in bronze of 110 MPa. (15)



(b) For the truss shown in figure below, determine the force in member AB, BC, GC, and FG. (20)

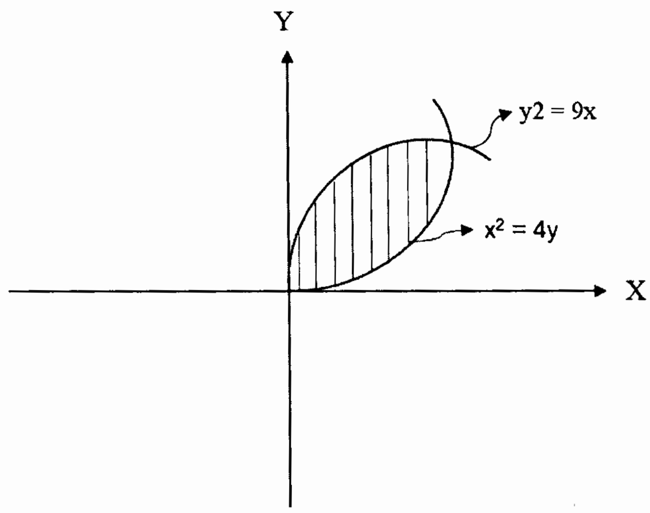


Section – B

5 (a) What is center of gravity and centroid of an area? (6)

(b) Find the centroid of a right circular cone, whose altitude is h and whose base has a radius r. (14)

(c) Determine the location of the centroid of the area enclosed by arcs of the parabolas $y^2 = 9x$ and $x^2 = 4y$, where the linear units are inches. (15)



6 (a) Define moment of inertia. Prove that $I_x = \bar{I}x + Ad^2$ where the symbols have their usual meaning. (12)

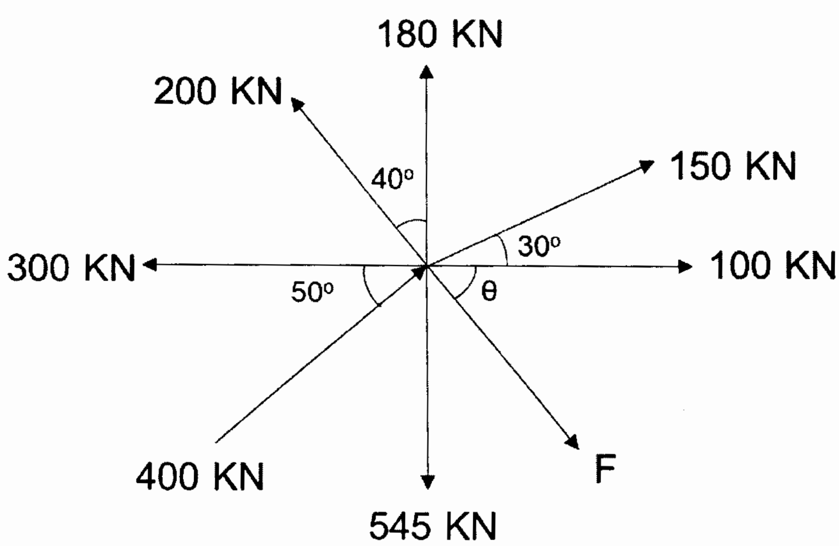
(b) Derive an expression for the moment of inertia of a rectangular area about axis through the central and parallel to a side. (10)

(c) Determine the moment of inertia of the steel section as shown in figure given below with respect to the x-axis and the centroidal axis parallel to x-axis. (13)

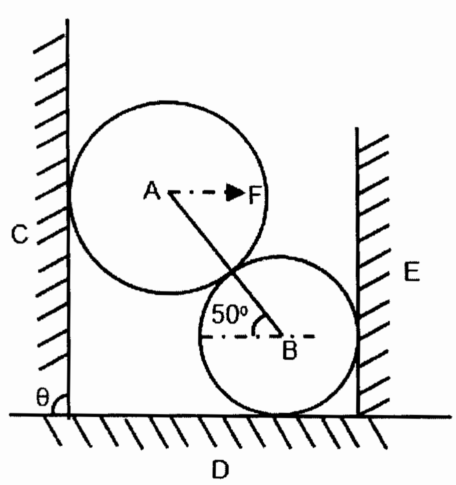
7 (a) Define the following terms: (09)

(i) Concurrent force system; (ii)Transmissibility of force system;
(iii) Free body diagram

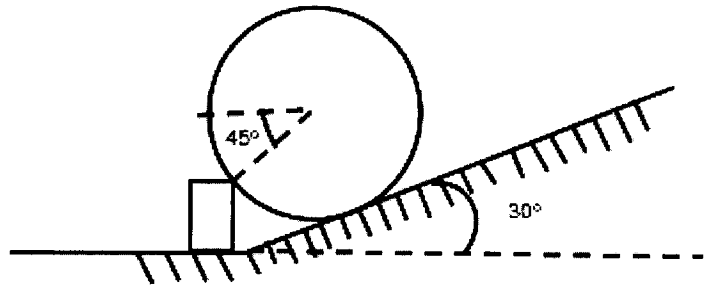
(b) Find the value of F and θ of the following force system shown in figure below. Also find the resultant of the force system. (11)



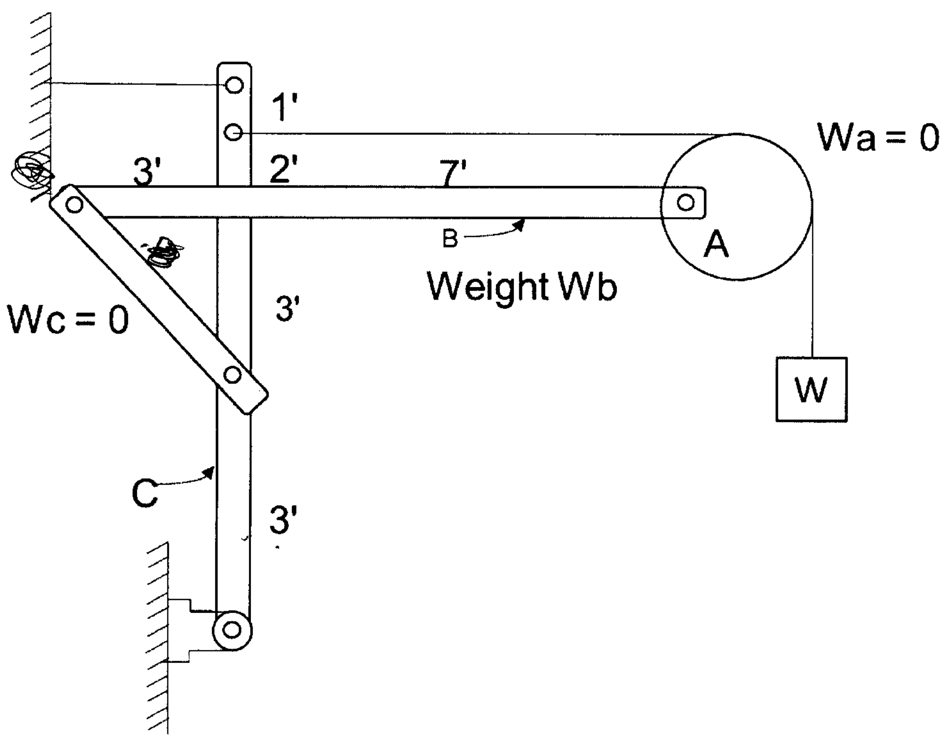
(c) Two spheres are at rest against smooth surfaces, as shown below. Sphere A weighs 3000 lb and sphere B weighs 500 lb. Let F = 1000 lb and θ = 90°. Find the reaction of C, D and E. (15)



8 (a) Draw the free body diagram of following figures. (10)



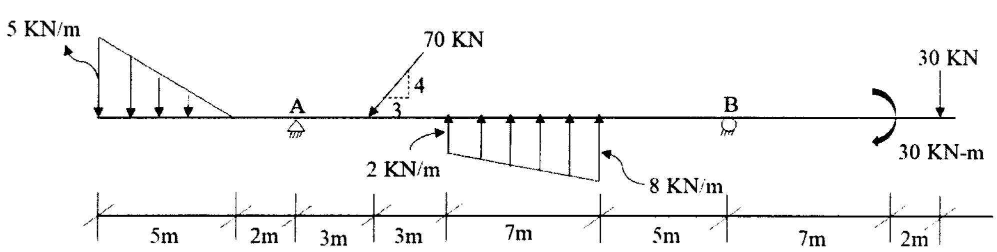
(i)



(ii)

(b) Describe the law of Parallelogram. (05)

(c) Determine the reactions at support A and B for the following beam subjected to the loading as shown in figure below. (12)



(d) What is force. Discuss briefly different types of force system. (08)

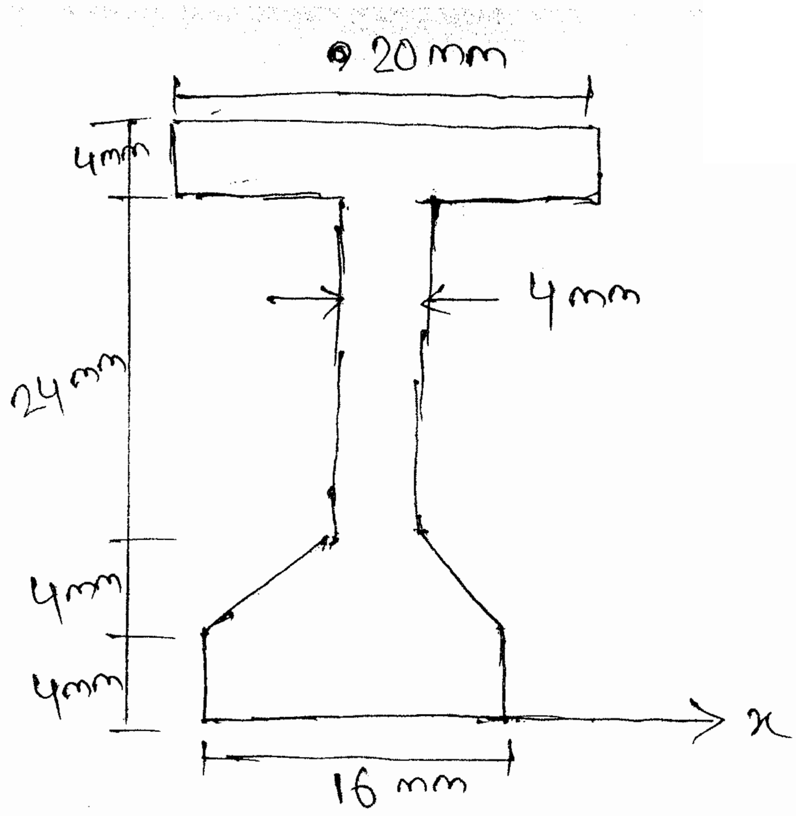


Figure for Question 6 (c)